



$$f(x) = x \sqrt{1-x^2}$$

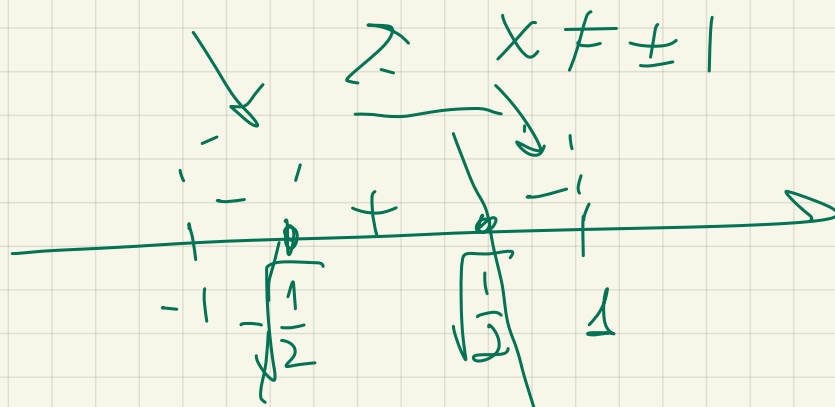
$$f'(x) = 1 \cdot \sqrt{1-x^2} + \frac{\cancel{x} \cdot 1 \cdot (-\cancel{2x})}{2\sqrt{1-x^2}} =$$

$$\frac{\sqrt{1-x^2} - x^2}{2\sqrt{1-x^2}} =$$

$$\frac{2(1-x^2) - x^2}{2\sqrt{1-x^2}} = \frac{1-x^2-x^2}{\sqrt{1-x^2}} = 0$$

$$\frac{1-2x^2}{\sqrt{1-x^2}} = 0$$

$$x^2 = \frac{1}{2}$$



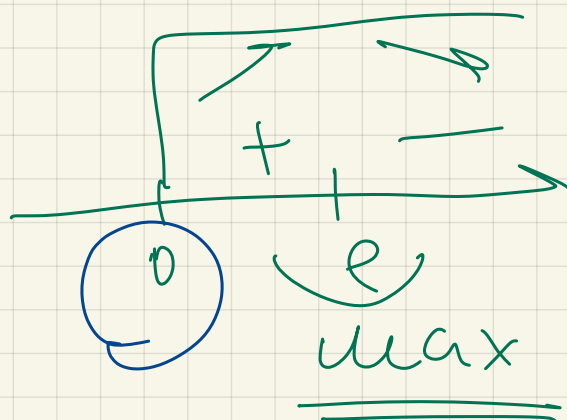
$$f(x) = \frac{\ln x}{x}$$

$$f'(x) = \frac{\frac{1}{x} \cdot x - 1 \cdot \ln x}{x^2} = \frac{1 - \ln x}{x^2} \quad x \neq 0$$

$$\frac{1 - \ln x}{x^2} = 0$$

$$\ln x = 1$$

$$x = e$$



$$f'(x) = 2e^{2x} - 4e^x + 1$$

$$e^x = t$$

$$2t^2 - 4t + 1 = 0$$

$$\frac{4 \pm \sqrt{16 - 2 \cdot 4}}{2} = \frac{1 \pm \sqrt{2}}{1} =$$

$$\frac{2 \pm \sqrt{2}}{2}$$

$$\ln e^x = \frac{2 + \sqrt{2}}{2}$$

$$e^x = \frac{2 - \sqrt{2}}{2}$$

$$x = \ln\left(\frac{2 + \sqrt{2}}{2}\right)$$

$$x = \ln\left(\frac{2 - \sqrt{2}}{2}\right)$$

$$\log_a b^c$$

$$f(x, y) = x^2 + xy + y^2 - 3x - 3y$$

$$\frac{\partial f}{\partial x} = 2x + y - 3$$

$$\frac{\partial f}{\partial y} = x + 2y - 3$$

$$\begin{cases} 2x + y - 3 = 0 \\ x + 2y - 3 = 0 \end{cases}$$

$$y = 3 - 2x$$

$$2(3-2x) + x - 3 = 0$$

$$6 - 4x + x - 3 = 0$$

$$3x = 3$$

$$|x| = 1$$

$$\{y = 1$$

$$\frac{\partial^2 f}{\partial x^2} = 2 = A$$

$$\frac{\partial^2 f}{\partial y^2} = 2 = c$$

$$\frac{\partial^2 f}{\partial x \partial y} = f = B$$

$$h = 4 - 1 = 3 > 0$$

$$A > 0$$

$\Rightarrow$ +.min

$$\beta = -4$$

$$f(x, y) = xy + \frac{1}{x} + \frac{1}{y}$$

$$x \neq 0$$

$$y \neq 0$$

$$\frac{\partial f}{\partial x} = y - \frac{1}{x^2} = 0$$

$$\frac{\partial f}{\partial y} = x - \frac{1}{y^2} = 0$$

$$y = \frac{1}{x^2}$$

$$x = \frac{1}{y^2} \Rightarrow x = \frac{1}{x^4}$$

$$x - \frac{1}{\left(\frac{1}{x^2}\right)^2} = x - x^4 = 0$$

$$x(1 - x^3) = 0$$

$$x^3 = 1$$

$$x = 0$$

$$\begin{cases} x = 1 \\ x = 0 \end{cases}$$

$$x = 1$$

$$y = 1$$

$$A = \frac{\partial^2 f}{\partial x^2} = \frac{2}{x^3}$$

$$A = 2$$

$$C = \frac{e}{y^2}$$

$$C = 2$$

$$B = 1$$

$$e^{x+y} = e^x \cdot \underbrace{e^y}$$

$$f(x,y) = e^{x+y} (x^2 - 2y^2)$$

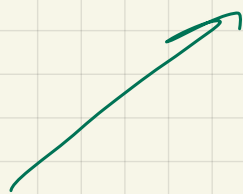
$$\frac{\partial f}{\partial x} = e^x \cdot e^y \cdot (x^2 - 2y^2) + 2x \cdot e^{x+y}$$

$$\frac{\partial f}{\partial y} = e^x \cdot e^y (x^2 - 2y^2) - 2 \cdot 2y \cdot (x^2 - 2y^2)$$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases}$$

$\Rightarrow$

$$\begin{cases} e^{x+y} (x^2 - 2y^2 + 2x) = 0 \\ e^{x+y} (x^2 - 2y^2 - 4y) = 0 \end{cases}$$



$$\begin{aligned} x^2 - 2y^2 + 2x - (x^2 - 2y^2 - 4y) \\ \cancel{x^2} - \cancel{2y^2} + 2x - \cancel{x^2} + \cancel{2y^2} + 4y = 0 \end{aligned}$$

$$2x + 4y = 0$$

$$\underline{x = -2y}$$



$$① \quad 4y^2 - 2y^2 - 4y = 0$$

$$2y^2 - 4y = 0$$

$$2y(y-2) = 0$$

$y = 0$ $x = 0$	$y = 2$ $x = -4$
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$$\left(l^{x+2} \right)' = l^{x+2}, \quad 1 = l^{x+2}$$

$$l^{2x+2} = l^{3x+2} \cdot 3$$

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= l^{x+y} (x^2 - 2y^2 + 2x) + l^{x+y} (2x+2) \\ &= l^{x+y} (x^2 - 2y^2 + 4x + 2) \end{aligned}$$

$$\frac{\partial^2 f}{\partial y^2} = e^{x+y} (x^2 - 2y^2 - 8y - 4) \quad \nwarrow$$

$$\frac{\partial^2 f}{\partial x \partial y} = e^{x+y} (x^2 - 2y^2 + 2x - 4y) \quad \swarrow$$

$$16 - 8 - 8 - 8 = -8$$

$$A = 2$$

$$B = 0$$

$$C = -4$$

$$D = -8 - 0 = -8 < 0$$

$$A > 0$$

крит. т.

не экстр

$$(-4; 2)$$

$$A = -6e^{-2} < 0$$

$$B = e^{-2} \cdot (-8)$$

$$D = 8 > 0$$

$$C = -12e^{-2}$$

$$-6 \cdot e^{-2} \cdot (-12)e^{-2} - (e^{-2}(-8))^2 =$$

$$\underline{72e^{-4}} - \underline{e^{-4} \cdot 64} =$$

$$e^{-4}(42-64) = 8$$

$$f(x,y) = \underbrace{e^{x+2y}}_{f(x)} \underbrace{(x^2-4y^2-2x+8y)}_{g(x)}$$

$$(f(x) \cdot g(x))' = f'(x) \cdot g(x) + g'(x) \cdot f(x)$$

$$\frac{\partial f}{\partial x} = e^{x+2y} (x^2-4y^2-2)$$

$$\frac{\partial f}{\partial x} = e^{x+2y} \cdot 2(x^2-4y^2-2x+8y) + e^{x+2y}(-8y+8)$$

$$= e^{x+2y} (2x^2-8y^2-4x+16y-8y+8)$$

$$e^{x+2y} (2x^2-8y^2-4x+8y+8)$$

$$2 \rightarrow (x^2 - 4y^2 - 2) = 0$$

$$2x^2 - 8y^2 - 4x + 8y - 8 = 0$$

$$-2 + 4x - 8y - 8 = 0$$

$$4x - 8y - 10 = 0$$

$$4x = 8y + 10$$

$$x = 2y + \frac{5}{2}$$

O/S

$$x =$$

$$y =$$

$$\frac{\partial^2 f}{\partial x^2}$$

$$\frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x \partial y}$$

